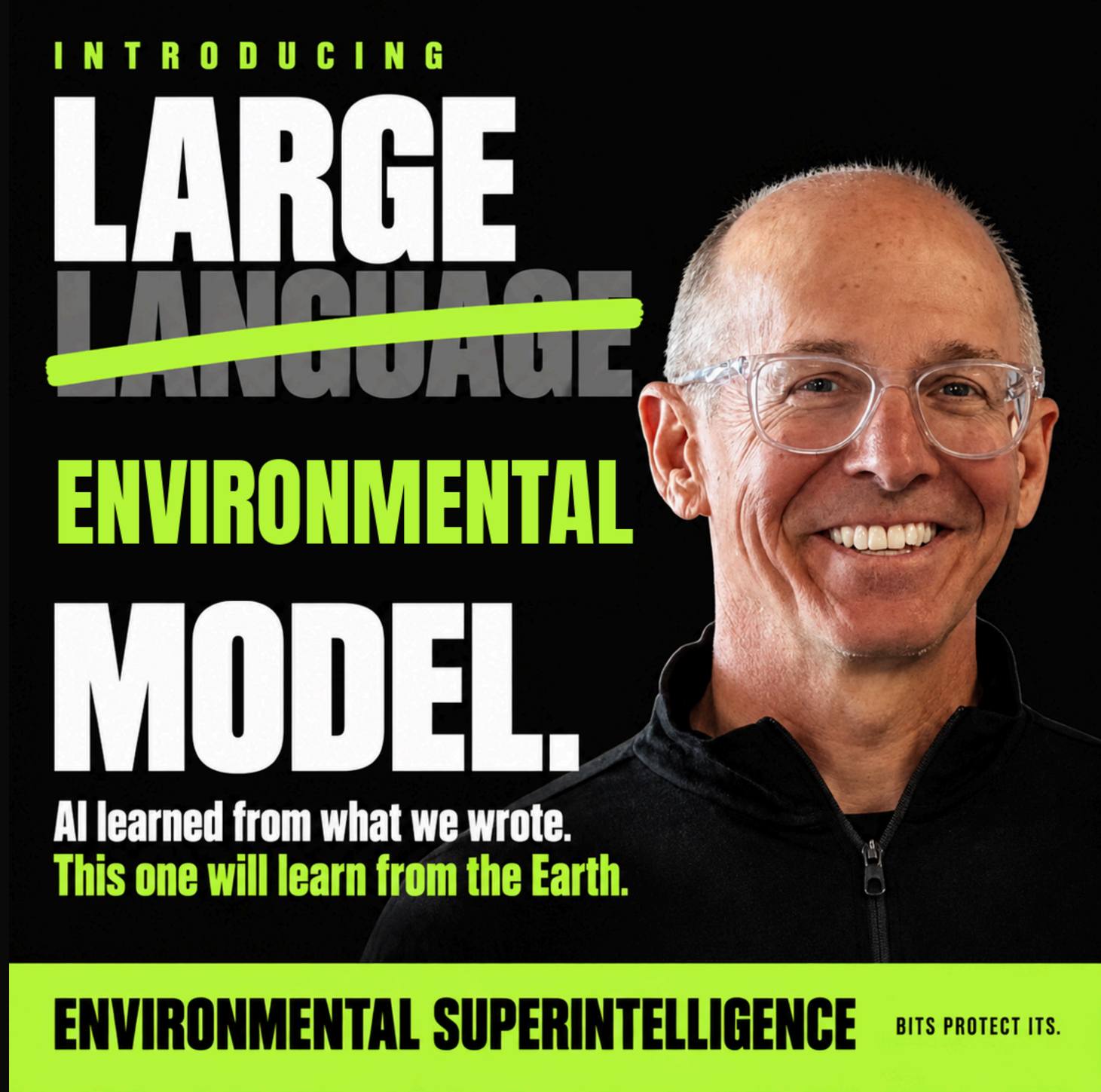
ENVIRONMENTAL SUPERINTELLIGENCE

INTELLIGENCE BUILT ON WHAT THE EARTH DOES

The AI labs trained on what humans wrote. The Large Environmental Model learns from Earth's living systems and is judged by one test: do they do better?

Jed Anderson · Founder and CEO, EnviroAI · October 2026



AI IS LEARNING FAST, FROM THE WRONG HALF OF THE WORLD

TRAINING COMPUTING

4 to 5×

growth a year, 2010 to 2024

Epoch AI

WHAT IT LEARNED FROM

Words

books, code, websites, forums

It can draft a permit. It cannot say what the river will do.

DEPENDS ON NATURE

$44T

more than half of world GDP

World Economic Forum, 2020

AI is now being put to work in the physical world: plants, grids, water, farms, coasts. What it learns, and what it is judged by, is being decided now.

Sources: Epoch AI, training compute of frontier models; World Economic Forum, New Nature Economy Report, 2020.

AI TRAINED ON NATURE ALREADY WINS

Model	Learned from	What it beat
AlphaFold	Nature's record of proteins and evolution	A 50-year problem in biology; Nobel Prize in Chemistry, 2024
GraphCast	Decades of Earth's weather	Europe's leading physics forecast on 90.3% of 1,380 measures
GenCast	The same record, as probabilities	Europe's ensemble forecast on 97.2% of 1,320 measures
Aurora	Over 1 million hours of Earth data	Matched or beat Europe's air pollution forecast on 74% of measures, about 50,000 times faster
AIFS	Earth's weather record	Made operational by ECMWF, February 2025, alongside its physics model

The pattern: learn from nature's own record, respect its physics, test in public, win. And it is cheap: GraphCast trained for about four weeks on 32 chips.

Sources: Nobel Prize; Google DeepMind (GraphCast 2023, GenCast 2024); Microsoft Aurora, Nature 2025; ECMWF.

THE OPEN LANE: NOBODY HAS JOINED THEM

Who	Learns from	Judged by nature doing better?
Frontier language models	What humans wrote	No
World-model labs (AMI Labs $1.03B, World Labs $1B, 2026)	Mostly video and simulation	No
Earth models (GraphCast, Aurora)	Weather, air, ocean; one domain each	No: forecast skill
Google Earth AI	Separate models wired by an agent; no biology, no law	No
Large Environmental Model	**Air, water, coast, land, life, and law, in one model**	**Yes: the Nature Benchmark**

A May 2026 survey of Earth-science foundation models describes unified, cross-sphere Earth models as an emerging goal, not yet achieved.

Sources: TechCrunch, March 2026; Yahoo Finance, February 2026; Google Research, October 2025; arXiv 2605.12542.

THE IDEA IN ONE LETTER

LANGUAGE MODELS LEARNED FROM WORDS. AN ENVIRONMENTAL MODEL LEARNS FROM THE EARTH.

Air monitors, river gauges, satellites, weather, soils, and species records, joined to the law that protects them.

LLM LEARNS FROM WORDS.

LEM LEARNS FROM EARTH.

ONE LETTER CHANGES WHAT AI LEARNS.

Large Language Model → Large Environmental Model



ENVIRONMENTAL SUPERINTELLIGENCE

BITS PROTECT ITS.

WHAT THE LARGE ENVIRONMENTAL MODEL IS

LANGUAGE CORE

READS THE LAW AND THE SCIENCE

An open-weight language model trained further on 22 million environmental documents. Explains and answers in plain words.

EARTH MODEL

PREDICTS WHAT NATURE DOES NEXT

Trained on air, water, coast, land, and life, with conservation of mass and energy built in. Predicts what an action would change.

ACTION LAYER

DOES THE WORK, WITH PEOPLE APPROVING

Permits, compliance, verification, and emergency modeling, learned from how the work is done and what happened after.

THE JUDGE The Nature Benchmark: open tests anyone can run. Does it predict nature better than physics models and frontier AI, and does it act safely for living systems?

A language model can read a permit but cannot predict a plume. A weather model can predict the plume but cannot read the permit. Protection takes both.

WHAT IT LEARNS FROM: THE EARTH'S RECORD

178.7 PB
NASA Earth data archive, growing about 160 TB a day

~4 billion
species observations in the global biodiversity record

12,165
USGS stream gauges measuring US rivers

22 million
permits and compliance documents already in EnviroAI

6,401
Texas permitted sites joined to the air monitors and rivers around them

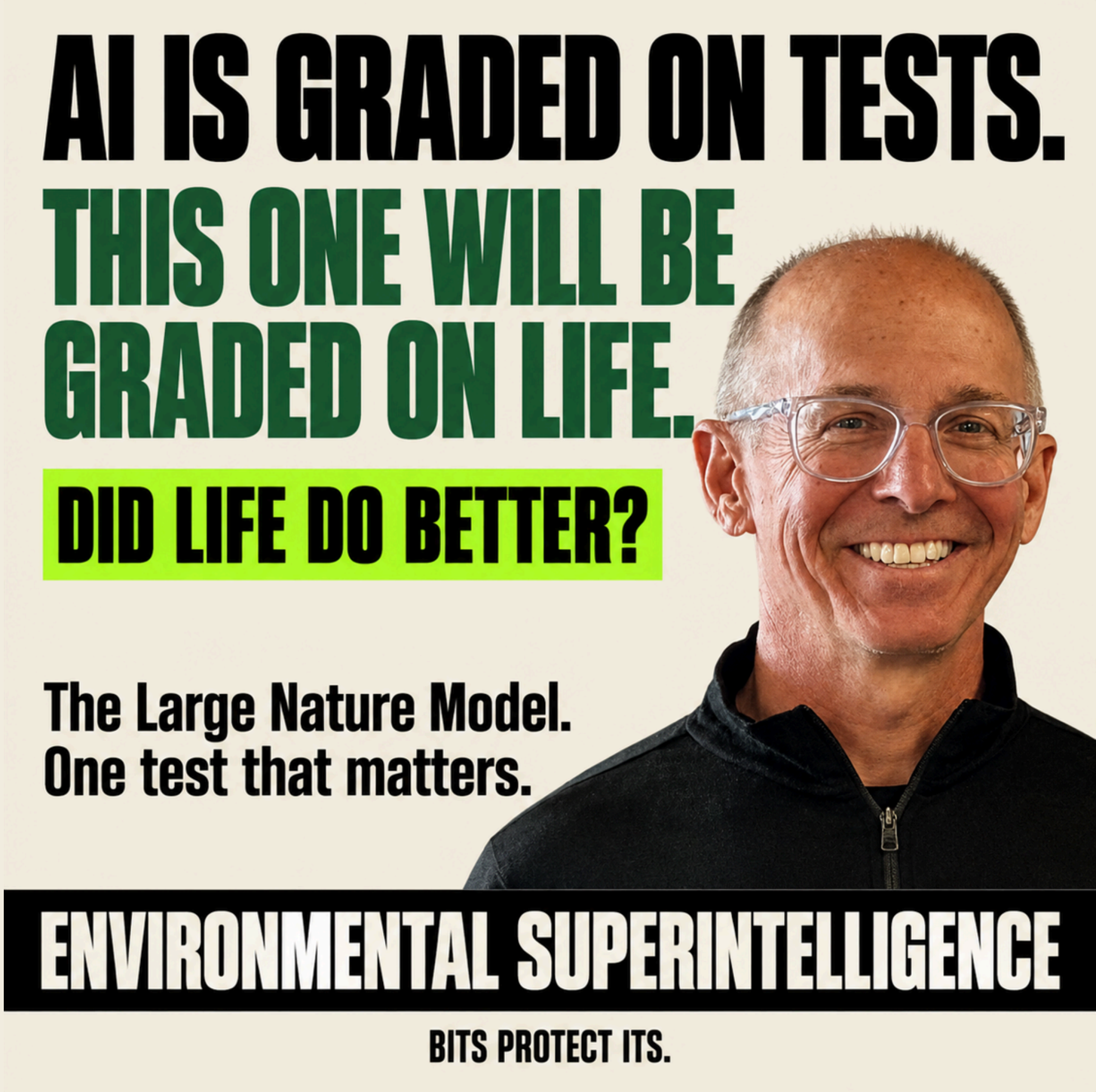
Every hour
new data from the deployed system, the record the next generation learns from

Sources: NASA Earthdata metrics FY2025; GBIF, October 2026; USGS 2024; EnviroAI. Green: public record. Cream: EnviroAI.

THE TEST THAT MATTERS

AI IS SCORED ON EXAMS. THIS ONE WILL BE SCORED ON OUTCOMES.

Did the air, the water, and the land do better? The Nature Benchmark makes that test open and public, for every AI model.



HOW IT IS JUDGED: THE NATURE BENCHMARK

An open, public test. The LEM is only better if it beats the best physics models and the best frontier AI on the tasks that protect living systems.

Test	Example task	Scored against
Predict	Streamflow at a Texas river gauge a week ahead; ozone downwind of the Houston Ship Channel tomorrow	What was measured
Trace	Find the source of a measured benzene spike from the permits, the wind, and the monitors	The confirmed source
Know the law	Which requirements apply to this unit, and is this permit application complete?	Agency decisions
Act safely	Propose an action and show what it does to the air, the water, and the habitat around it	The Environmental Safety Bench

Tasks shown are examples of the kind; the benchmark's tasks are set with university partners and published in full.

THE BUILD: THREE GENERATIONS

1 · MONTHS 0 TO 9

LEARN THE RECORD

Language core trained on 22 million documents; first Earth models for Texas air and water; benchmark version 1 published.

256 to 512 GPUs

under 1 MW · about $4M to $14M

2 · MONTHS 9 TO 18

MODEL THE LIVING WORLD

One Earth model, 10 to 50 billion parameters, across air, water, coast, land, and life, joined to the language core.

2,000 to 4,000 GPUs

3 to 6 MW · about $33M to $105M

3 · MONTHS 18 TO 36

FRONTIER SCALE

Continuous training at the scale of the leading labs, on the whole Texas record and every place that joins.

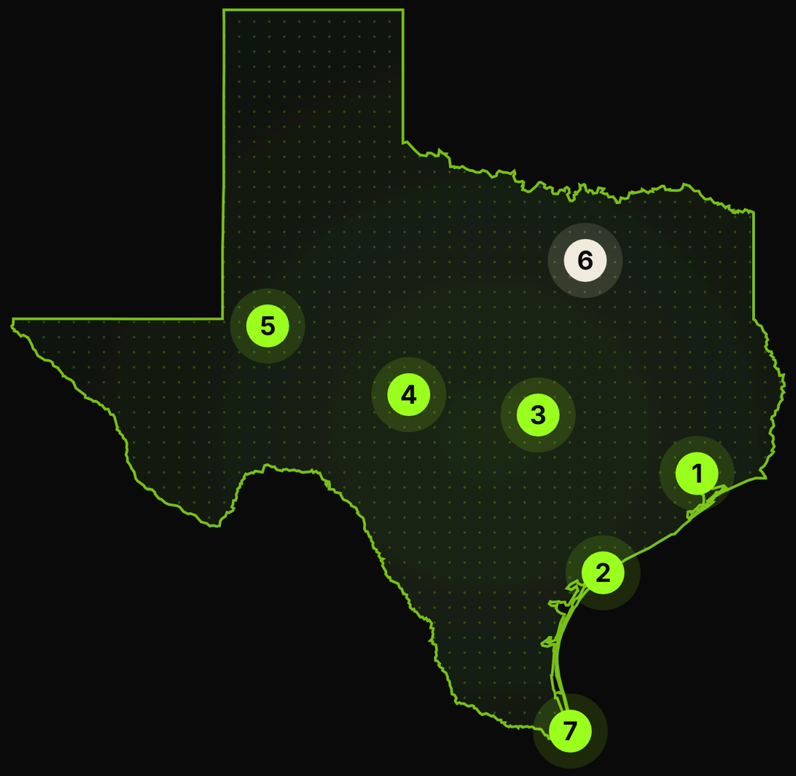
Up to 57,000 GPUs

up to 80 MW · the Texas 1% compact

Generations 1 and 2 are affordable now on rented computing and private capital. Generation 3 needs tens of megawatts nobody can rent cheaply. That is what the compact supplies.

Costs: GPUs running nine months at $2.50 to $4.00 per GPU hour (2026); power at 1.4 kW per GPU including cooling.

WHY TEXAS: SEVEN LIVING SYSTEMS AND THE COMPUTING



1 · Industry and air. The nation's largest refining and chemical corridor; 6,401 permitted sites joined to their air monitors

2 · The Gulf coast. Bays, marshes, and barrier islands

3 · Water. 23 river basins and the aquifers beneath them

4 · Land and habitat. Ranchland, forests, and wildlife

5 · Carbon. Carbon stored in rock and soil

6 · The computing. Data centers; 1% is about 80 MW, 57,000 GPUs

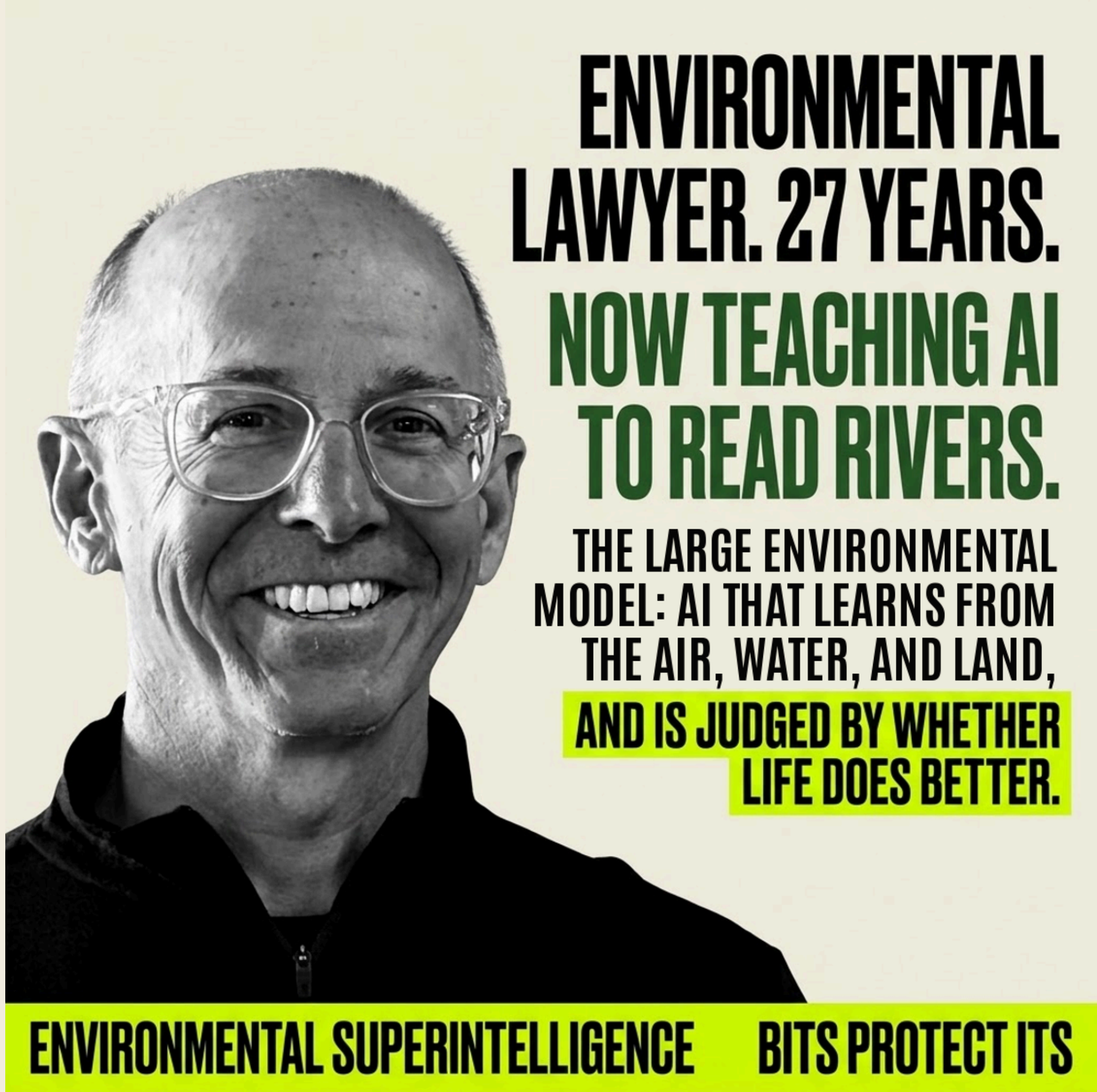
7 · A launch site. Boca Chica, on a coast rich with wildlife

Locations are general. Compact arithmetic: 1% of about 8 GW of 2025 Texas data center peak demand, at 1.4 kW per GPU.

WHO IS BUILDING IT

NINE YEARS BUILDING ENVIRONMENTAL AI. NOW, THE LEAP.

Year	Milestone
2018	First environmental AI tool
2019	First Fortune 500 customer
2022	Environmentally trained AI assistant
2025	AI agent prepares a complete environmental permit
2026	**The Large Environmental Model**



WHO WINS

Party	What they get	What they give
Texas business	Permitting and compliance run on a model that knows both the law and the physics; foresight on water and storms	Their hardest problems as test tasks
Texans	Their air, water, coast, and land predicted and protected, with the test results public	Nothing
Texas universities	Leadership of a field that does not exist yet; research computing; co-authorship	Researchers and data
State agencies	A model their staff use free for public work, trained on their own record	Data sharing
The AI field	An open test of whether AI is safe and useful for living systems	Their models, run against it
Nature	For the first time, an intelligence judged by whether living systems do better	Nothing; it has given enough

WHAT WE ASK, IN THE NEXT 90 DAYS

Who	The ask
Texas universities	Join as research partners in the model and the Nature Benchmark
TCEQ, TWDB, GLO	Open data partnerships so the model learns from the state's full environmental record
Industry	Become founding partners; bring the problems that matter most; run it in your operations first
Investors and computing providers	Fund Generations 1 and 2, which are affordable now
The Governor	Announce the Texas 1% Compute Compact, which carries the model to frontier scale

What EnviroAI brings on day one: the language layer in use with industrial customers, 22 million documents, 6,401 Texas sites joined to the air monitors and rivers around them, continuous air and water modeling in demonstration, and the first Environmental Safety Bench.

TEXAS CAN BUILD IT FIRST.

The computing, the industry, and the living systems are here. The starting layer is built. What it takes now is the decision to go big.

Onward. Upward.

Jed Anderson · jed@enviro.ai · enviro.ai

THE LARGE NATURE MODEL

"They built AI on what we wrote." *I'm building it on what the Earth does."*



JED ANDERSON
Founder and CEO, EnviroAI

ENVIRONMENTAL SUPERINTELLIGENCE